Activity - 3
Section – 3J
Date – 11-Sept-25

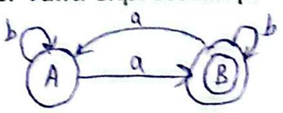
Student Roll No. _ _
Student Name:
Total marks: 20

Q:1 - Given an FA for language **accepting Odd Number of a's where ∑ = {a, b}. (3+3 marks)**

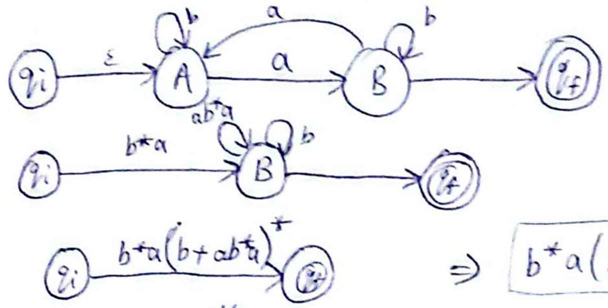
a) **You have to Eliminate states one at a time to get** ***b*a(b + ab*a)**** and

b) **Another valid expression** ***(b + ab*a)**** ab*

DFA :



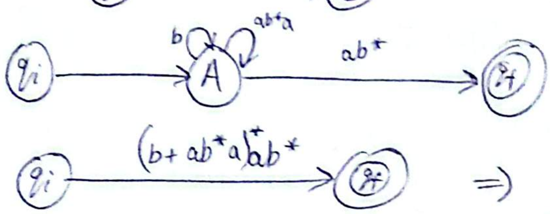
(a)



(First eliminating A then B)

$\Rightarrow$ $\boxed{b^*a(b+ab^*a)^*}$

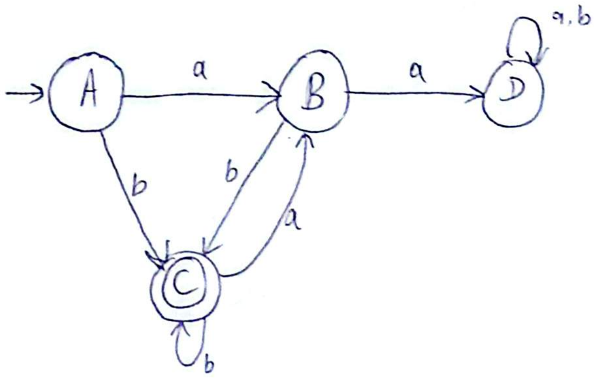
(b)

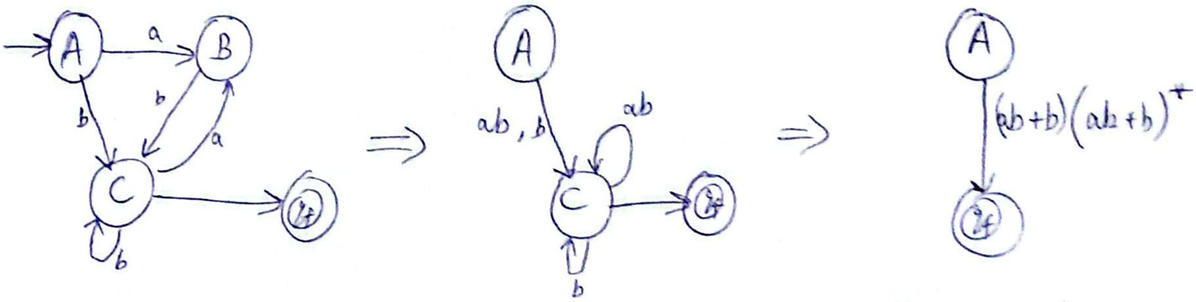


(First eliminating B then A)

$\Rightarrow$ $\boxed{(b+ab^*a)^*ab^*}$

Q:2 - Make FA state diagram for language that accepts strings **Ending with b and Not Containing** aa where ∑ = {a, b} **and get Regular expression for the language by State Elimination. (4 marks)**





$\therefore (ab+b)(ab+b)^* = (ab+b)^+$

Q:3 - Make FA state diagram where $\Sigma = \{a, b\}$ with **Regular** expression as

**(aa+bb+(ab+ba)(aa+bb)*(ab+ba))*** **(10 marks)**